Square-Mile-Size AI Data Centers Requiring GW-scale Power Connections

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PRESENTATION FLOW

- Snapshot of GW Scale Data Centers
- Load Characteristics of AI Data Centers
- Impact of AI Data Centers on Grid

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Snapshot of GW Scale Data Centers

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Snapshot of GW Scale Data Centers

S. No	Project / Company	Location	Approx Site Area (sq mi)	Approx Area (sq km)	Approx Capacity (GW)
1	Fermi America	Wyoming / Texas, USA	9	23	10
2	Meta Hyperion	Holly Ridge, Louisiana, USA	4.25	11	5
3	OpenAI Stargate	Abilene, Texas, USA	1.35	3.5	5
4	Adani–Google AI Campus	Visakhapatnam, India	0.77	2	1
5	Reliance–Digital Connexion	Visakhapatnam, India	0.62	1.6	1
6	YEIDA AI Supercomputing Hub	Greater Noida, India	0.45	1.2	1

Fermi America's proposed U.S. data center campus spans about one-third the size of Chandigarh.

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Load Characteristics of AI Data Centers

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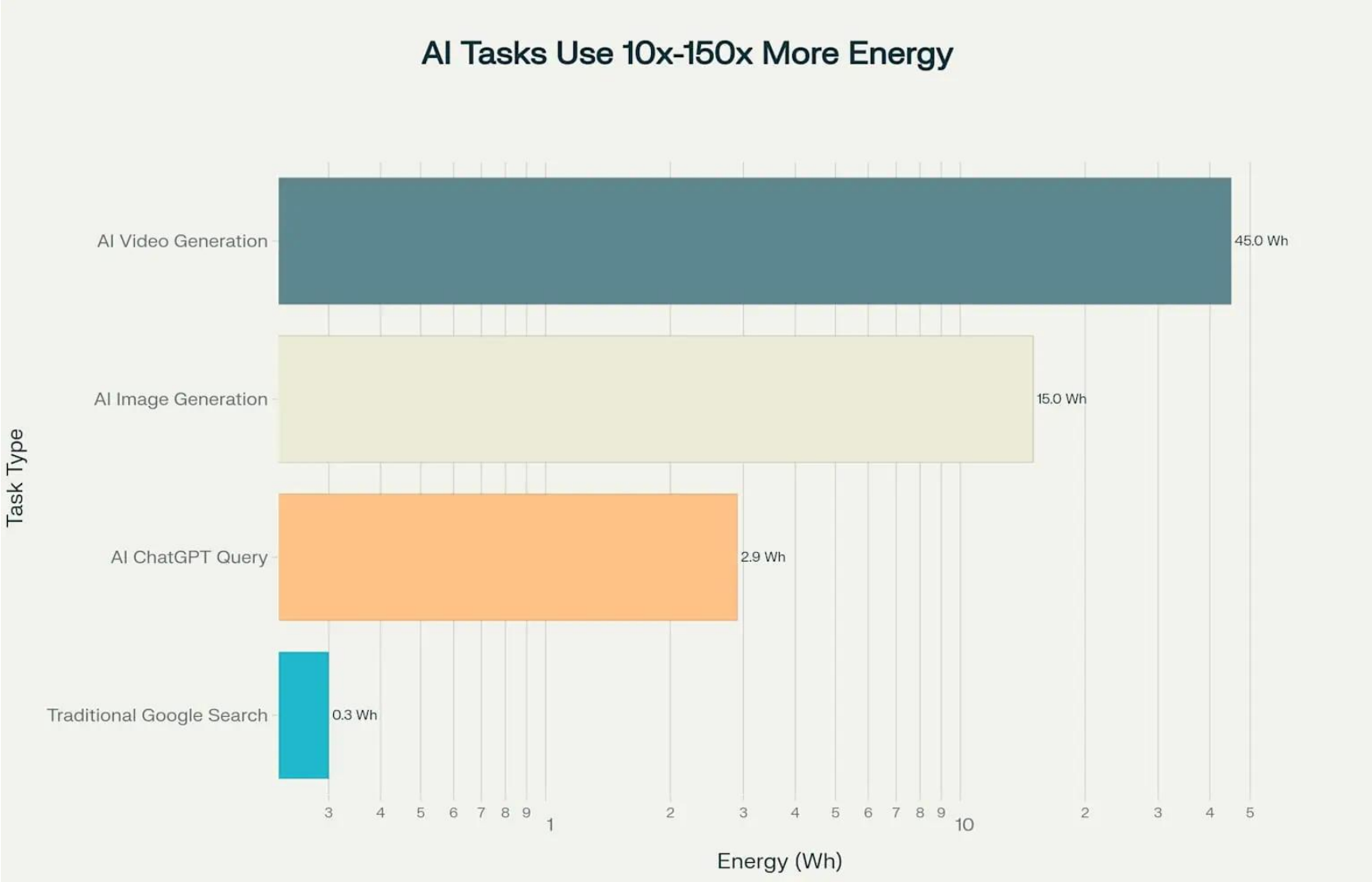
Load Characteristics – AI Data Centre



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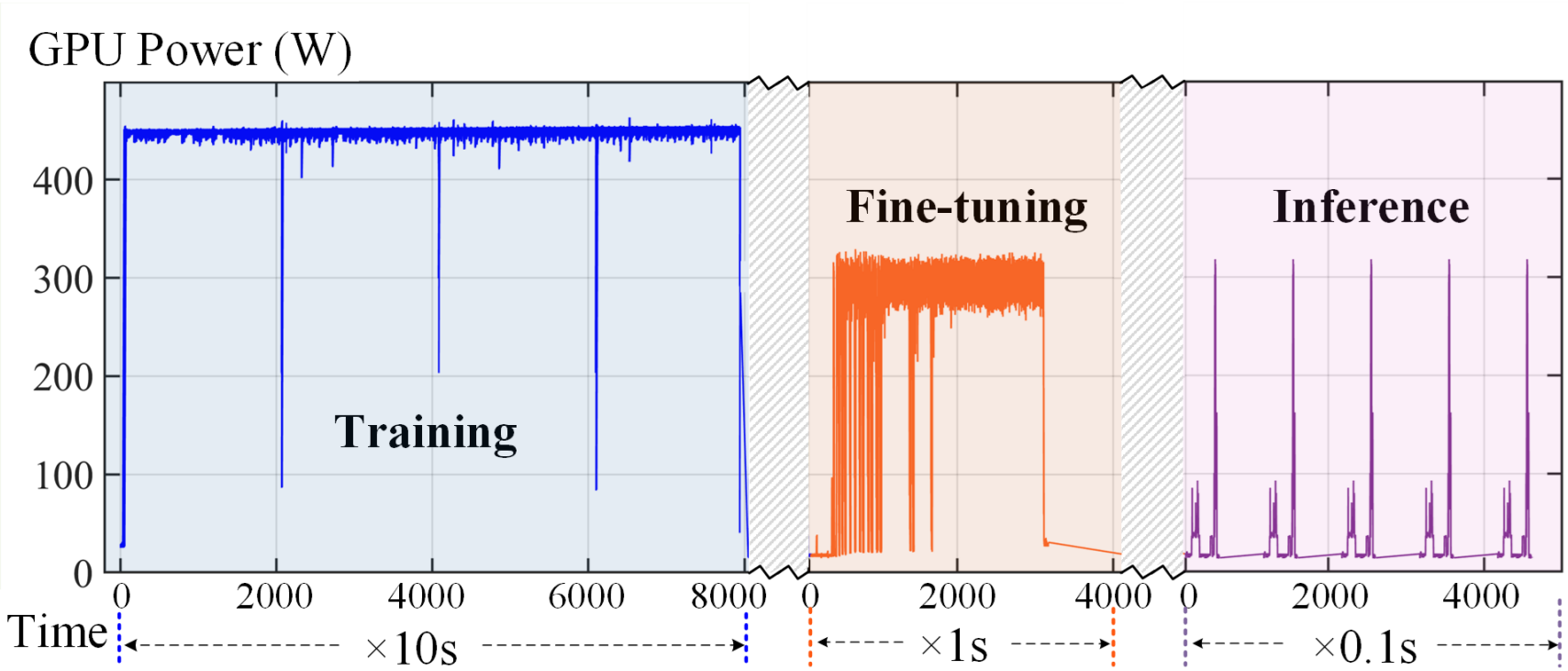
Load Characteristics – AI Data Centre



Load Characteristic During Various Phases of Operation of AI – Huge Variations

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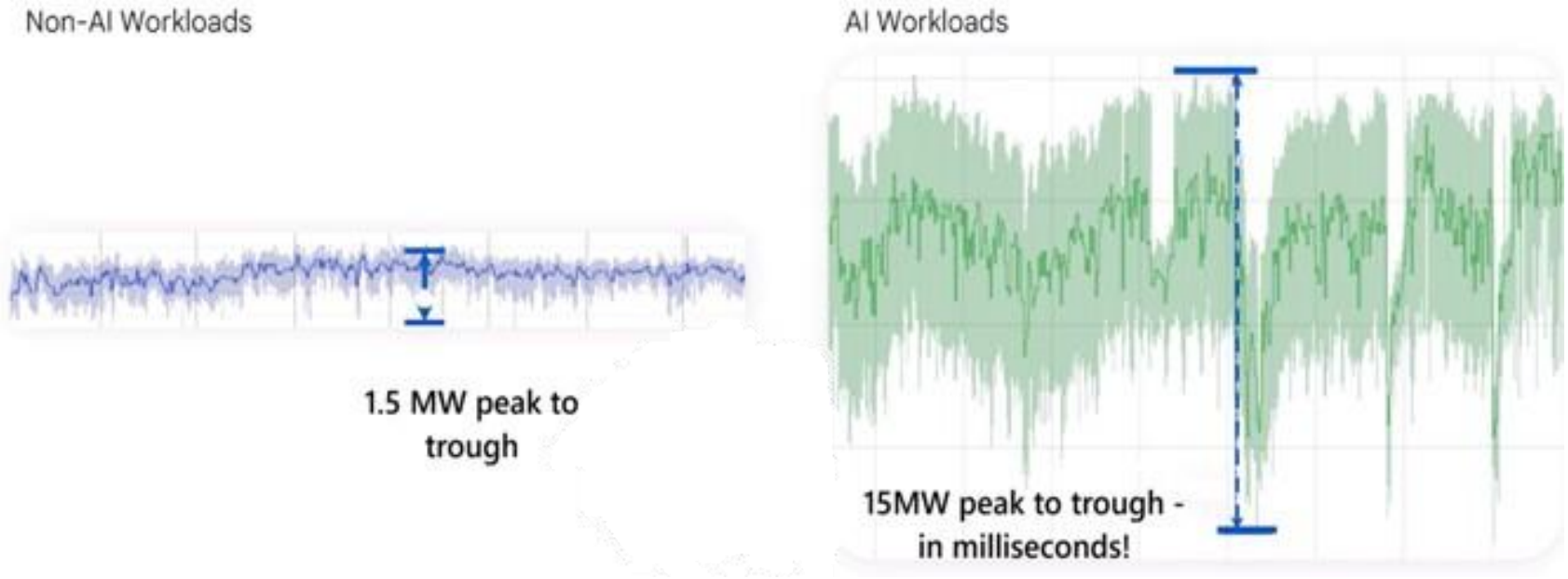
Load Characteristics – Non AI Vs AI Data Centre



Aspect	Traditional	AI Workloads
Power Density	10-15 kW/rack	30+ kW/rack
Load Variability	Stable, predictable	Highly variable, bursty
Computational Intensity	Moderate	Extremely high
Data Dependency	Moderate	High
Phase Change	Rare	Training/Fine-tuning/Inference

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Impact of AI Data Centers on Grid

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Impact on Grid

Real Time Impact

- Frequency Variation
- Voltage Variation
- Power Quality Issues (THD, Flicker)

Short Term Impact

- Load Forecasting Issues

Long Term Impact

- Grid Capacity Constraints
- Power Quality Issues (Equipment Ageing)

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Frequency Variation (Real Time Impact)

Load Characteristic

- Sub-second to seconds-scale fluctuations

Impact on Grid

- Fossil fuel generators have ramp rates of 5-50 MW/minute
- Managing hundreds of MWs variation in fraction of a second is not feasible
- Power-angle oscillations in nearby generators.
- Frequency changes with high rate of change of frequency.
- Risk of underfrequency load shedding or system trips.

Mitigation Strategy:

- Fast Frequency Response (FFR): Deploy on site grid-scale batteries
- AI load modulation: Control of compute cycles in sync with grid frequency

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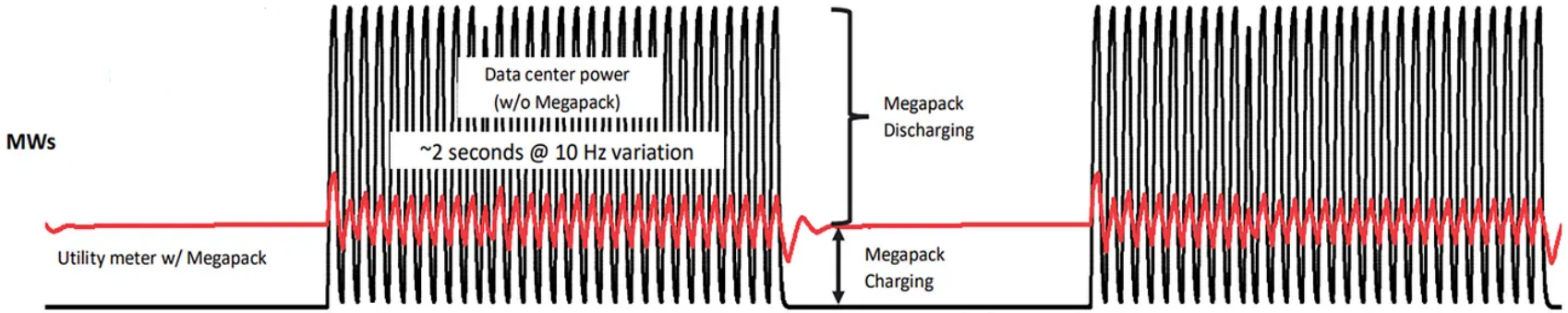
Frequency Variation - Mitigation Through BESS



Comparison of Data Centre Load Curve – Without BESS (Black) Vs With BESS (Red)

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Voltage Variation (Real Time Impact)

Load Characteristic

- AI Data Centers Contain large UPS systems and cooling compressors
- Fast reactive power swings when clusters activate simultaneously.

Impact on Grid

- Traditional tap changers or capacitor banks -too slow for real-time dips.
- Inability to comply with voltage regulation standards
- Degraded power delivery to nearby industrial or sensitive customers.
- Possible stalling of motors or tripping of voltage-sensitive equipment.

Mitigation Strategy:

- Install STATCOMs/SVCs for fast dynamic voltage support

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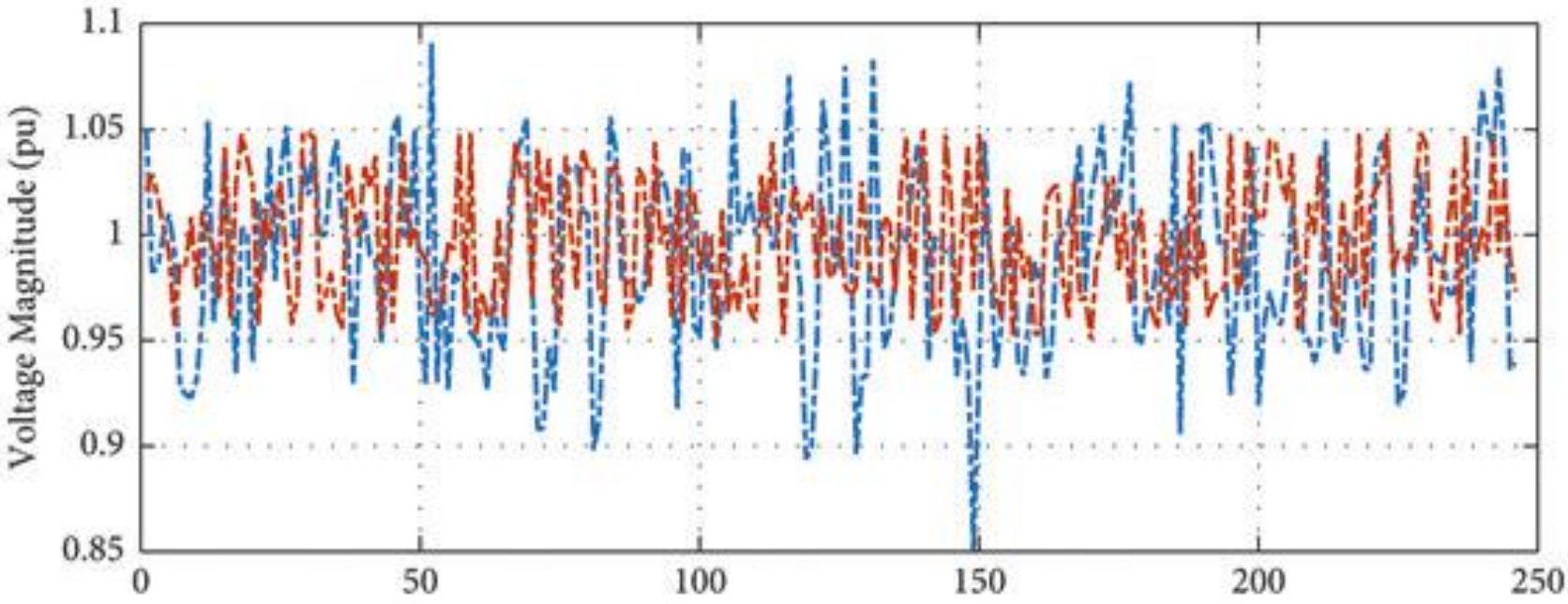
Voltage Variation - Mitigation Through STATCOM



Voltage Profile - Pre Vs Post Optimal STATCOM Placement

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Power Quality Issues (Real Time/Long Term Impact)

Load Characteristic

- Thousands of SMPS and UPS units introduce current harmonics.

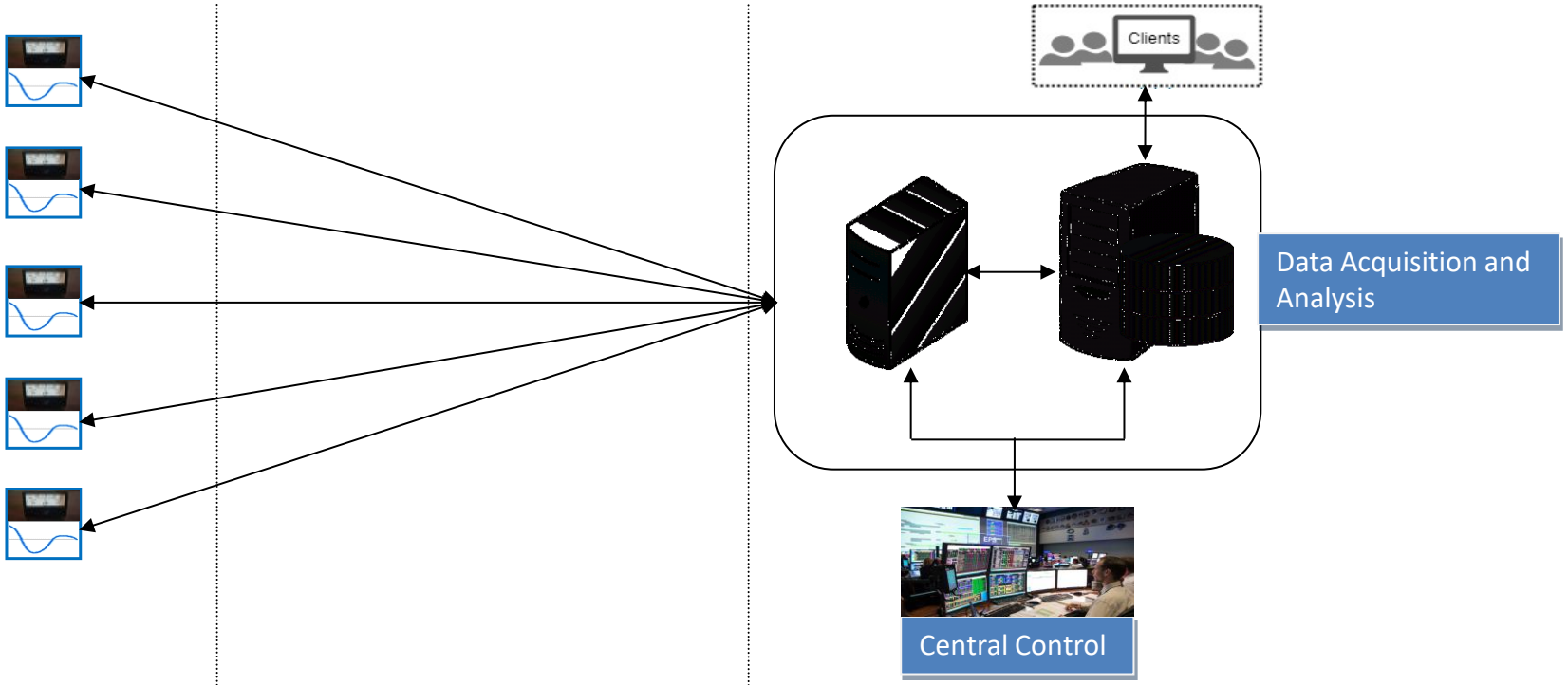
Impact on Grid

- Increased Total Harmonic Distortion (Real time)
- Maloperation of protection relays (Real time)
- Flicker and electromagnetic interference (Real time)
- Overheating of equipment and insulation stress (Long term)
- Accelerated ageing and reduced equipment lifespan (Long term)

Mitigation Strategy

- Use of active harmonic filters at load end
- Enforce Power Quality monitoring

Power Quality Issues – Mitigation Through PQ Monitoring



PQ Meters

Communication Protocol- IEC 61850 and Medium- Ethernet/ cellular (4G)

Enterprise Level Data Acquisition and Management Software

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Load Forecasting Issues (Short Term Impact)

Load Characteristic

- Workload-driven, random
- Load uncertainty

Impact on Grid

- Forecasting models not yet trained on AI demand patterns.
- Uncertain forecasts challenge dispatch planning.
- Stress on generation ramping and reserves.

Mitigation Strategy

- Use machine-learning-based load forecasting tuned to AI load profiles.
- Set connection ramp-rate limits in grid agreements.
- Integrate on-site renewables and BESS for smoothing.
- Implement demand response with flexible job scheduling.

Grid Capacity Constraints (Long Term Impact)

Load Characteristic

- GW scale demand

Impact on Grid

- Network Capacity Challenges
- Equipment overload
- Lack of adequate redundancy

Mitigation Strategy

- Strategic siting near substations
- Network Upgrades
- Hybrid renewable + storage projects
- Long-term interconnection planning

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Thanks

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